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SOC

Analyst L1 For Free

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Malwares

What is Malware?

Malicious Software designed to:

- Steal data
- Disrupt systems
- Gain unauthorized access
- Spy on users
- Encrypt files for ransom

Types of Malwares



- Virus
- Worm
- Trojan
- Ransomware
- Spyware
- Rootkit

Types of Malwares



Virus

A virus attaches itself to a legitimate file and spreads when that file runs.

How It Works:

1. User downloads infected file
2. File runs
3. Virus executes
4. Spreads to other files

Example:

An infected .exe shared via USB.

SOC Detection:

- Unknown process spawning
- Suspicious file modification
- Antivirus alerts



Worm

Definition:

A worm spreads automatically over networks, no user action needed.

Key Difference:

Virus → Needs user interaction

Worm → Self-propagates

Famous Example:

WannaCry (spread using SMB vulnerability)

SOC Detection:

- Sudden network traffic spike
- Same payload across multiple hosts
- Lateral scanning behavior



Trojan

Definition:

Malware disguised as legitimate software.

Example:

Fake “Free Cracked Software.exe”

What It Does:

- Opens backdoor
- Steals data
- Installs additional malware

SOC Detection:

- Unusual outbound traffic
- Unknown services running
- Suspicious startup entries



Ransomware

Definition:

Encrypts victim files and demands ransom payment.

Attack Flow:

1. Phishing email
2. User clicks attachment
3. Malware installs
4. Files encrypted
5. Ransom note displayed

SOC Indicators:

- Mass file renaming
- High CPU usage
- Shadow copy deletion
- Unusual encryption activity



Spyware

Definition:

Secretly collects user information.

What It Steals:

- Passwords
- Banking info
- Screenshots
- Keystrokes (Keylogger)

SOC Detection:

- Strange outbound connections
- DNS queries to unknown domains
- Data exfiltration alerts



Rootkit

Definition:

Malware designed to hide other malware.

Dangerous Because:

- Hides processes
- Hides files
- Hides registry entries
- Very difficult to detect

SOC Detection:

- Mismatch between user-mode and kernel-mode results
- Integrity check failures
- Unusual driver loads

Types of Malwares



Comparison

Type	Needs User?	Spreads Automatically?	Primary Goal
Virus	Yes	No	Infect files
Worm	No	Yes	Spread rapidly
Trojan	Yes	No	Backdoor
Ransomware	Yes	Sometimes	Encrypt files
Spyware	Yes	No	Steal data
Rootkit	No	No	Hide malware



SOC Analyst Mindset

When you see an alert, ask:

- Is this malware delivery?
- Is it persistence?
- Is it lateral movement?
- Is data being exfiltrated?
- Is this encryption behavior?

SOC L1 is about:

- Detect
- Validate
- Escalate